

The hire graph of the 3-free door

Let χ_3 be the non-principal Dirichlet character modulo 3: $\chi_3(n) = 0$ if $3 \mid n$, and $\chi_3(n) = \left(\frac{n}{3}\right)$ otherwise, so $\chi_3(n) = +1$ if $n \equiv 1 \pmod{3}$ and $\chi_3(n) = -1$ if $n \equiv 2 \pmod{3}$. For an odd prime $p \neq 3$ write

$$m_0(p) = p + \chi_3(p), \quad m_1(p) = p - \chi_3(p).$$

Lemma 1. $m_0(p)$ is even and not divisible by 3. $m_1(p)$ is divisible by 6. $\{m_0(p), m_1(p)\} = \{p-1, p+1\}$. In particular 3 never divides a 3-free door.

Proof. The residues of $p-1, p, p+1$ modulo 3 are a permutation of $\{0, 1, 2\}$, and $p \not\equiv 0 \pmod{3}$. Hence exactly one neighbour of p is divisible by 3. Both neighbours are even. If $p \equiv 1 \pmod{3}$ then $\chi_3(p) = +1$, so $m_0(p) = p+1 \not\equiv 0 \pmod{3}$ and $m_1(p) = p-1 \equiv 0 \pmod{3}$. If $p \equiv 2 \pmod{3}$ then $\chi_3(p) = -1$, so $m_0(p) = p-1 \not\equiv 0 \pmod{3}$ and $m_1(p) = p+1 \equiv 0 \pmod{3}$. In both cases $m_1(p)$ is even and divisible by 3, hence by 6. $\square$

Examples. $p = 11 \equiv 2 \pmod{3}$, so $\chi_3(11) = -1$, $m_0(11) = 10 = 2 \cdot 5$, $m_1(11) = 12 = 2^2 \cdot 3$. $p = 13 \equiv 1 \pmod{3}$, so $\chi_3(13) = +1$, $m_0(13) = 14 = 2 \cdot 7$, $m_1(13) = 12 = 2^2 \cdot 3$.

Lemma 2. The four residue classes of odd primes modulo 12 determine the doors as follows.

$p \pmod{12}$	χ_3	$m_0 \pmod{12}$	$v_2(m_0)$	$m_1 \pmod{12}$	$v_2(m_1)$
1	+1	2	= 1	0	≥ 2
5	-1	4	≥ 2	6	= 1
7	+1	8	≥ 2	6	= 1
11	-1	10	= 1	0	≥ 2

Proof. The classes $\{1, 5, 7, 11\} \pmod{12}$ are the units modulo 12. Each line is the conjunction of Lemma 1 with the unique lifting of that class modulo 4 and modulo 3. Residues 2 and 10 modulo 12 are $2 \pmod{4}$, so $v_2(m_0) = 1$; residues 4 and 8 are $0 \pmod{4}$, so $v_2(m_0) \geq 2$. $\square$

Lemma 3. If p and $p+2$ are both prime and $p > 3$, then $p \equiv 2 \pmod{3}$ and

$$m_1(p) = m_1(p+2) = p+1.$$

Proof. The pair of residues of $(p, p+2)$ modulo 3 is either $(1, 0)$ or $(2, 1)$. The first makes $p+2$ divisible by 3, impossible for $p+2 > 3$. Thus $p \equiv 2 \pmod{3}$ and $p+2 \equiv 1 \pmod{3}$, so $\chi_3(p) = -1$ and $\chi_3(p+2) = +1$. Then $m_1(p) = p+1$ and $m_1(p+2) = p+1$. $\square$

Corollary 1. If an even integer is not divisible by 3, at most one of its two neighbours is an odd prime greater than 3.

Lemma 4. If p and $q = 2p+1$ are both prime and $p > 3$, then $p \equiv 2 \pmod{3}$ and $m_0(q) = 2p$.

Proof. If $p \equiv 1 \pmod{3}$ then $q = 2p+1 \equiv 0 \pmod{3}$, impossible for $q > 3$. Thus $p \equiv 2 \pmod{3}$, so $q \equiv 2 \pmod{3}$, $\chi_3(q) = -1$, and $m_0(q) = q-1 = 2p$. $\square$

The hire graph. An odd prime $p \neq 3$ is an *owner* of an odd prime q when $q \mid m_0(p)$. By Dirichlet's theorem on the two admissible classes modulo $3q$,

$$p \equiv 1 \pmod{3}, \quad p \equiv -1 \pmod{q} \quad \text{and} \quad p \equiv 2 \pmod{3}, \quad p \equiv 1 \pmod{q},$$

every odd prime $q \neq 3$ has an owner. Hence the set of primes that occur as 2 or as an odd prime factor of some $m_0(p)$ is all primes except 3.

Spectra are taken on finite windows. For $X \geq 5$, let the owners be the odd primes $p \neq 3$ with $p \leq X$. Let S_X consist of 2 together with every odd prime factor of $m_0(p)$ for such owners, and write $n = |S_X|$. The *hire graph* G_X has vertex set S_X , a spoke from 2 to every other vertex, and a directed gold arc $p \rightarrow q$ whenever $p \leq X$, $p, q \in S_X \setminus \{2\}$, and $q \mid m_0(p)$ (gold witnessed by owners in the window). The underlying undirected gold graph has an edge $\{p, q\}$ when at least one of the two arcs is present. Let H_X be the combinatorial Laplacian of the undirected graph (spokes plus undirected gold), and let L'_X be the Laplacian of the undirected gold graph on $S_X \setminus \{2\}$.

The relation $q \mid m_0(p)$ is not symmetric: $5 \mid m_0(11) = 10$ while $11 \nmid m_0(5) = 4$.

Lemma 5. *There is a gold arc $p \rightarrow q$ if and only if $p \equiv -\chi_3(p) \pmod{q}$.*

Proof. The congruence is $q \mid (p + \chi_3(p))$, i.e. $q \mid m_0(p)$. □

Lemma 6. *Every undirected gold edge $\{p, q\}$ is the base of a triangle 2– p – q in G_X .*

Proof. Both p and q are adjacent to 2 by construction. □

Lemma 7. $\lambda_{\max}(H_X) = n$. *The vector with value $n - 1$ at 2 and value -1 at every other vertex is an eigenvector for this eigenvalue, independently of the gold edges.*

Proof. Let v be that vector. On the star alone, $H_X v = n v$. Each gold edge has both ends in the set where v is constant, so contributes 0 to $H_X v$. □

Lemma 8. *Let $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_{n-1}$ be the eigenvalues of L'_X . Then*

$$\text{spec}(H_X) = \{0, n\} \cup \{1 + \mu_2, \dots, 1 + \mu_{n-1}\}.$$

In particular $\lambda_2(H_X) = 1$ if and only if the undirected gold graph on $S_X \setminus \{2\}$ is disconnected.

Proof. Order the vertices as 2 followed by $S_X \setminus \{2\}$. Then

$$H_X = \begin{pmatrix} k & -\mathbf{1}^\top \\ -\mathbf{1} & I + L'_X \end{pmatrix},$$

where $k = n - 1$. Vectors constant on $S_X \setminus \{2\}$ yield eigenvalues 0 and n . On the complement $\mathbf{1}^\top y = 0$ the lower block acts as $1 + \mu$, and the copy of $\ker L'_X$ spanned by $\mathbf{1}$ has already been used. The remaining eigenvalues of L'_X are $\mu_2, \dots, \mu_{n-1}$. □

On the windows below, undirected gold remains disconnected, so $\lambda_2(H_X) = 1$.

X	$ S_X $	gold edges	gold arcs	L'_X connected	$\mu_2(L'_X)$	$\lambda_2(H_X)$
30	4	1	1	no	0	1
50	6	3	3	no	0	1
100	12	7	7	no	0	1
200	18	13	13	no	0	1

$X = 30$: $S = \{2, 5, 7, 11\}$, gold $\{5, 11\}$. $X = 50$: $S = \{2, 5, 7, 11, 19, 23\}$, gold $\{5, 11\}$, $\{5, 19\}$, $\{11, 23\}$. $X = 100$: gold $\{5, 11\}$, $\{5, 19\}$, $\{5, 41\}$, $\{7, 13\}$, $\{7, 29\}$, $\{11, 23\}$, $\{19, 37\}$.

Variant: the sluice. The hire graph G_X uses m_0 throughout. A finite set Σ of odd primes defines a variant door: the Σ -door of p is $m_0(p)$ if no prime of Σ divides $m_0(p)$; otherwise $m_1(p)$ if no prime of Σ divides $m_1(p)$; and is otherwise undefined.

Lemma 9. *If $\Sigma = \{5\}$, the Σ -door is defined for every odd prime $p \neq 3$. It differs from $m_0(p)$ if and only if $p \equiv 11$ or $19 \pmod{30}$.*

Proof. The condition $5 \mid m_0(p)$ is $p \equiv -\chi_3(p) \pmod{5}$. Combined with p odd and the value of $\chi_3(p)$, this is the pair of classes $11, 19 \pmod{30}$. On those classes $m_1(p)$ is divisible by 6 and not by 5: the two neighbours of p differ by 2, so at most one is divisible by 5. $\square$

Lemma 10. *If $\Sigma = \{5, 7\}$, the Σ -door is undefined if and only if*

$$p \equiv 29, 41, 169, \text{ or } 181 \pmod{210}.$$

These are four residue classes modulo 210, of density $4/\varphi(210) = 1/12$ among units modulo 210.

Proof. The door is undefined precisely when 5 divides one neighbour of p and 7 divides the other, i.e.

$$\{p-1, p+1\} \equiv \{0 \pmod{5}, 0 \pmod{7}\}$$

as sets. The system $p \equiv \pm 1 \pmod{5}$, $p \equiv \mp 1 \pmod{7}$, together with p odd, cuts four classes modulo $210 = 2 \cdot 3 \cdot 5 \cdot 7$. The odd representatives are 29, 41, $169 \equiv -41$, and $181 \equiv -29$. There are $\varphi(210) = 48$ units modulo 210, so the density among those units is $4/48 = 1/12$. $\square$